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Behavioral Economics Series 1: Prospect Theory

Prospect Theory evaluates how individuals choose under risk and uncertainty, and aims to illustrate that sometimes choices are not optimal. This theory was first developed in 1979 by two psychologists, Amos Tversky and Daniel Kahneman, and a lot of behavioural

economists later built on their work. In 2002, Professor Kahneman received the Nobel Prize in Economics for his work in Behavioural Economics (unfortunately, professor Tversky already passed away in 1996, and the Nobel Prize doesn't award posthumously).

While the essays published by KT (Tversky and Kahneman) involves a wide array of ideas, they can be mostly summarized into two integral effects: the certainty effect and the reflection effect.

1. The Certainty Effect

Consider two different investment options. Option 1 provides a return of \$3000 with no risk at all (the returns are completely certain). Option 2 an 80% chance to return \$4000 and a 20% chance of no return. From an expected value (or expected utility) perspective, a rational person would take option 2 (which has an expected return of \$3200) over option 1 (which has an expected return of \$3000). However, through a series of experiments, KT discovered that under such circumstances, people were more likely to choose option 1, despite the lower expected return. KT concluded that people are risk-averse when it comes to potential gains. In other words, people prefer certain gains over uncertain gains.

2. The Reflection Effect

Now consider a choice among potential losses. You must choose between (A). a 100% chance of losing \$3000 and (B): an 80% chance of losing \$4000. Again, in terms of expected value, a rational person would prefer option A (which gives an expected loss of -\$3000) to option B (which gives an expected loss of -\$3200). Yet KT's experiments showed that most people would choose option B. That is, people are risk-seeking when it comes to potential losses, as they would gamble on

option B simply because it offers the possibility of no loss, despite the higher expected loss. People really hate losses, and this emotional effect against losses, known as risk aversion, is the reason why people tend to choose option B.

As you can see, the numbers in the two examples are essentially the same. However, in the first example, people choose the option with the lower expected value (expected gain), yet people tend to choose the option with a higher expected value (expected loss). People are risk-averse when facing possible gains yet they become risk-seeking when facing possible losses. This inconsistency is why this effect is called the reflection effect.

Example 1:	Option 1	Option 2
Potential Gains	100% chance to gain \$3000 (risk-averse)	80% chance to gain \$4000
Example 2:	Option A	Option B
Potential Losses	100% chance to lose \$3000	80% chance to lose \$4000 (risk-seeking)

The certainty effect and the reflection effect can explain a lot of observations in the financial markets. For instance, people tend to sell “winning” stocks prematurely, which can be explained by the certainty effect. Selling early to secure gains is analogous to option 1 in the first example (100% to gain \$3000) and holding on to the winning stock is similar as option 2 (80% chance to gain more but a 20% chance of having nothing). As most people chose the suboptimal option in KT’s experiment, many investors in real life tend to sell early to lock gains. Similarly, people also tend to hold on to losing stocks for way too long. This is similar to the examples of potential losses in the reflection effect. People tend to hold onto losing stocks for too long. Holding onto losing stocks is analogous to option B in example 2 (80% chance to lose more but 20% chance of no loss), which investors prefer to cutting losses and lose a certain amount (which is similar to option A). In general, prospect theory helps explain a lot of investors’ behaviour in finance and illustrates that people are not necessarily consistent or rational when facing potential losses or gains

It is also worth noting that, in the past three decades since prospect theory was developed, its main application has mainly been limited to the financial markets. The key reason why it’s rarely applied outside of finance is mainly because it is hard to know how and where to apply it, as a scholar from Yale has pointed out. I myself do have a few suggestions for possible areas to apply prospect theory. First of all, I think perhaps the psychological effects revealed by the certainty and reflection effects could be applied to the field of game theory, especially the area of contracts and negotiations where economic agents are usually assumed to be rational. Another possible application might be in the area of consumer choice. Since the utility gained from consumption is analogous to the returns generated by investors, we could infer that consumers would also choose certain utility over uncertain utility. For instance, say you saw online that there is a restaurant that is “very good” (let’s say, 80% chance of getting 50 utils) yet you always go to a restaurant is “quite good” (let’s say, 38 utils). In certain cases, you might just stick with the “quite good” restaurant

because you know for certain how much pleasure you will get. Of course, these are just some postulations that might be interesting to think about, as I don't really know for certain whether or not prospect theory would be applicable to these fields.

References:

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